

Project Proposal

COURSE CODE: CSE 4204 || COURSE TITLE: MOBILE COMPUTING LAB

Semester: Fall 2026

Project Title: NUBTK_COPILOT - AI-Powered University ERP and Career Architect

1. Team Information

Name	Student ID	Role & Responsibility
Bashir Alam	11220320963	Django backend, Celery automation, OpenRouter API, ML models, System architecture.
Tisha Rani Chakroborty	11220320954	Frontend. HTML and Tailwind CSS.
Fariha Rahman Oishee	11220320967	Database Manager. PostgreSQL tables in Supabase
Ratul Raha Shuvo	11220320949	Backend Support. Payment gateway, and API testing.

Instructor: Md. Riaz Mahmud, Assistant Professor, Dept of CSE, NUBTK

Section: 8D | **Team ID:** CSE4204-8D-T01

2. Problem Statement

Current university management systems are limited to administrative data entry like results and attendance. There is no granular tracking of practical skills that students acquire after each semester, such as 'Python Loops' or 'SQL Joins'. As a result, final-year students cannot identify if their skills match job market demands. Additionally, faculty members lack data-driven insights to identify which students are struggling in specific subjects. NUBTK_COPILOT aims to solve this gap by bridging academia and industry requirements.

3. Objectives

- To develop a smart ERP system that tracks students' academic records along with technical skill inventory.
- To build an AI Mentor Hub for programming, career guidance, and skill improvement through chat.

- To automate administrative workflows including admission, payment, and weekly reports.
- To provide a 'Career Readiness Score' based on CGPA and market data (BDJobs/LinkedIn).

4. Proposed Features

Feature	Description
Skill Inventory Module	Students log skills (e.g., SQL Joins) after courses for future reference.
AI Mentor Hub	OpenRouter API (Claude 3) provides coding help and career advice with gamified points.
Admin Insight Engine	Celery sends automated weekly reports on student performance to admins.
Career Readiness Score	Calculated (0-100) based on CGPA + Skills. Provides feedback for job readiness.

5. Technology Stack

Component	Technology	Justification
Backend	Django 5.0, Python 3.11	Rapid development, high security, and robust API support.
Database	PostgreSQL (Supabase)	Real-time capabilities and large dataset handling.
AI/ML	OpenRouter API, Scikit-learn	Claude 3 for mentoring, ML for performance prediction.
Frontend	HTML, Tailwind CSS	Lightweight and fully responsive design.

6. Expected Outcome

NUBTK will receive a fully automated system where students identify skill gaps through AI mentorship and faculty get proactive notifications. This serves as a portfolio project demonstrating full-stack, AI, and automation expertise.

GitHub Repository: <https://github.com/bashirnubtk/cse4204-8d-t01-nubtk-copilot>